

Metropolitan University, Sylhet
Department of Computer Science and Engineering
Summer Semester Final Examination – 2024
Program: B.Sc. in CSE Batch: 58
Course: GED 431: Business Communication

Time: 2 Hours

Marks: 40

Answer any **four** questions. All parts of each question must be answered sequentially. Figures in the right margin indicate full marks.

1. (a) You are about to develop an advertisement for a multinational software company; in this process, you have to consider some cultural variables. Now, identify the variables that require assessment. 5
- (b) If you have identified the necessary variables, develop a cultural matrix to plan your best communication strategy. 5
2. (a) "The act of integrity makes a person more charismatic than actually having integrity." What is your opinion regarding the statement? 5
- (b) On the basis of your knowledge of "the principle of liking" in the science of persuasion, build an argument. Supporting the statement-"People tend to respond or comply in a scenario where they find similarities. 5
3. (a) You represent a data analytics company. Your assignment is to give a presentation on the new processing interface at A2i for seed funding. Outline the uses of your Charismatic Leadership Tactics (CLT) to get maximum attention from the participants. 5
- (b) Can "positional power" be bypassed by "expertise?" Develop your answer with an example. 5
4. (a) In your experience, can you illustrate how an individual can become a charismatic leader by just engaging in conversation? 5
- (b) According to "the principle of social proof", people follow the lead of similar others; give a discussion of the statement with your agreement or disagreement. 5

5. (a) "In the influence game, emotion is like playing with fire". Give two examples, one for how emotion can be a strong influence tool and another for how it can impact the communication process negatively. 5
- (b) Reciprocity is the game changer for a new business relations establishment—justify the statement on the basis of "the principle of reciprocity." 5
6. (a) For modern business communication, story-telling is a must. Now point out what makes a good story. 5
- (b) In your experience, do you think an individual with authority has a better persuasion changes rather than someone who does not, build your argument. 5

Metropolitan University
 Department of Computer Science and Engineering
 Summer Semester Final Examination – 2024
Program: B.Sc. in CSE; Batch: 58
Course: MAT 216: Geometry and Vector Analysis

Time: 2 Hours

Marks: 40

Answer any **four** questions sequentially. All parts of each question must be answered sequentially. Figures in the right margin indicate full marks.

1. a) Define scalar and vector product of two vectors $\vec{A}$ and $\vec{B}$. 2
 b) Illustrate the geometrical interpretation of dot and cross product of two vectors $\vec{A}$ and $\vec{B}$. 4
 c) Find the area of parallelogram having sides of vectors are $\vec{A} = 2\hat{i} - 3\hat{j} + 4\hat{k}$ and $\vec{B} = \hat{i} + 2\hat{j} - \hat{k}$. 4

2. a) Prove that $\vec{A} \cdot (\vec{B} \times \vec{C}) = \vec{B} \cdot (\vec{C} \times \vec{A}) = \vec{C} \cdot (\vec{A} \times \vec{B})$ 5
 b) Find the volume of the parallelepiped whose three sides of vectors are $2\hat{i} - \hat{j} + 2\hat{k}$, $\hat{i} + 2\hat{j} + 2\hat{k}$, $3\hat{i} - 4\hat{j} + 5\hat{k}$. 5

3. a) If $\vec{A}$ has constant magnitude, then show that $\vec{A}$ and $\frac{d\vec{A}}{dt}$ are perpendicular provided $\left| \frac{d\vec{A}}{dt} \right| \neq 0$. 2.5
 b) A particle moves along the curve $x = 2t^2$, $y = t^2 - 4t$, $z = 3t - 5$, where t is the time. Find the components of its velocity and acceleration at time $t = 5$ in the direction $\hat{i} - 3\hat{j} + 2\hat{k}$. 5
 c) If $\phi(x, y, z) = x^2y^2$ and $\vec{A} = x^3\hat{i} + y^2\hat{j} + z^3\hat{k}$ then find $\frac{\partial^3(\phi\vec{A})}{\partial x^2 \partial z}$. 2.5

4. a) Define directional derivative. 1
 b) Find the directional derivative of $\phi = x^2yz + 4xz^2$ at $(1, -2, -1)$ in the direction $2\hat{i} - \hat{j} + 2\hat{k}$. 4
 c) Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at the point $(2, -1, 2)$. 5

5. Evaluate $\iint \vec{A} \cdot \hat{n} dS$, where $\vec{A} = 18z\hat{i} - 12\hat{j} + 3y\hat{k}$ and S is that part of the plane $2x + 3y + 6z = 12$ which is located in the first octant. 10

6. a) Define direction cosines and direction ratios. 2
 b) If the coordinates of the points P and Q are respectively $(2, 3, 5)$ and $(-1, 3, 2)$, find the direction cosines of the line PQ . 4
 c) If l , m and n are the direction cosines of a line, then prove that $l^2 + m^2 + n^2 = 1$. 4

Metropolitan University
Department of Computer Science and Engineering
Summer Semester Final Examination – 2024

Program: B.Sc. in CSE; Batch: 58
Course: CSE 211: Digital Logic Design

Time: 2 Hours

Marks: 40

Answer any **four** questions. All parts of each question must be answered sequentially. Figures in the right margin indicate full marks.

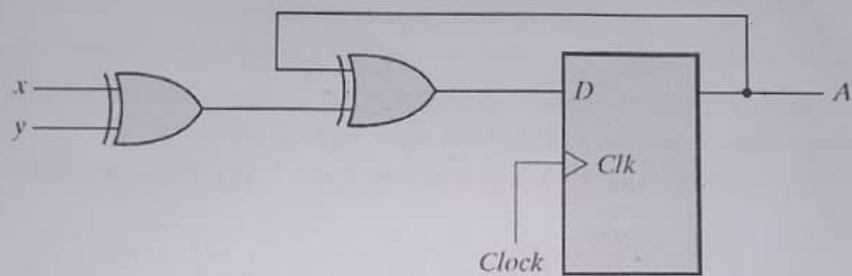
1. (a) Convert the decimal value 3720 in each of the following number systems (you must show calculation). (i) binary (ii) hex (iii) octal (iv) BCD 5
(b) Simplify the following Boolean functions T_1 and T_2 to a minimum number of literals: 5

A	B	C	T_1	T_2
0	0	0	1	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	0	1
1	0	1	0	1
1	1	0	0	1
1	1	1	0	1

2. (a) Using Karnaugh Map Simplify the following Boolean function F, together with the don't-care conditions d. 5
 $F(A, B, C, D) = \sum(5, 6, 7, 12, 14)$
 $d(A, B, C, D) = \sum(3, 9, 11, 15)$
(b) Explain how a 4-bit (Binary Adder) parallel adder works. Illustrate your explanation with an example of adding 1011 and 0101. 5
3. (a) Implement the following Boolean function with a Multiplexer. 5
 $F(A, B, C, D) = \sum(1, 2, 4, 5, 6, 9, 11, 13, 15)$
(b) Construct a 16x1 multiplexer using 4x1 multiplexers showing block diagrams with appropriate labeling and connection. 5
4. (a) Design a logic circuit that controls an elevator door in a three-story building. M is a logic signal that indicates when the elevator is moving ($M = 1$) or stopped ($M = 0$). F1, F2, and F3 are floor indicator signals that are normally LOW, and they go HIGH only when the elevator is positioned at the level of that particular floor. For example, when the elevator is lined up level with the second floor, $F2 = 1$ and $F1 = F3 = 0$. The circuit output is the **DOOR** signal which is normally LOW and is to go HIGH when the elevator door is to be opened. Find the simplified **DOOR** function and draw the circuit. 6
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- (b) Prove the universality of NAND and NOR gate. 4

PTO

- 5 (a) Design a 3 to 8 line decoder circuit. 3
- (b) Using 8-bit binary representation, perform the signed addition of these two decimal numbers: +45 and -25. 2
- (c) Design a four-input priority encoder. 5
6. (a) Show the operation of a basic SR Latch circuit. 4
- (b) Derive the state table and the state diagram of the sequential circuit shown below. 6



Metropolitan University
Department of Computer Science and Engineering
Summer Semester Final Examination - 2024
Program: B.Sc. in CSE; Batch: CSE 58
Course: CSE 231: Algorithm Design and Analysis

Time: 2 Hours

Marks: 40

Answer any **four** sets of questions. All parts of each question must be answered sequentially. Figures in the right margin indicate full marks.

1. Three matrices are given with their dimensions:

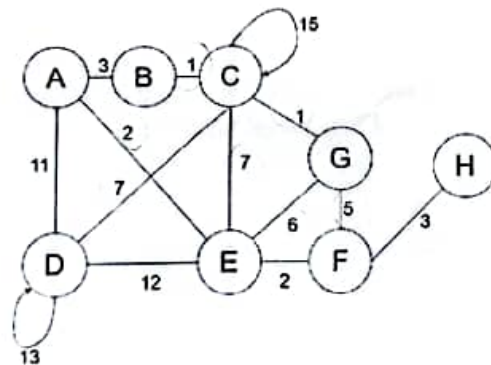
$$A_1 = 2 \times 6, \quad A_2 = 6 \times 7, \quad A_3 = 7 \times 3 \quad \& \quad A_4 = 3 \times 9$$

(10)

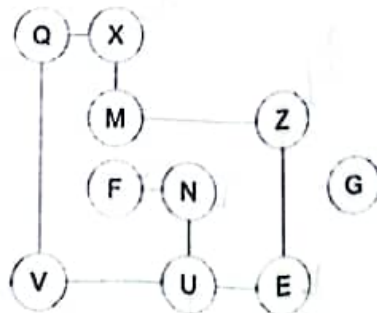
Using the dynamic programming approach for Matrix Chain Multiplication,

- i) Find minimum number of multiplications required for multiplying all matrices.
- ii) Find the Optimal Order of Multiplication.

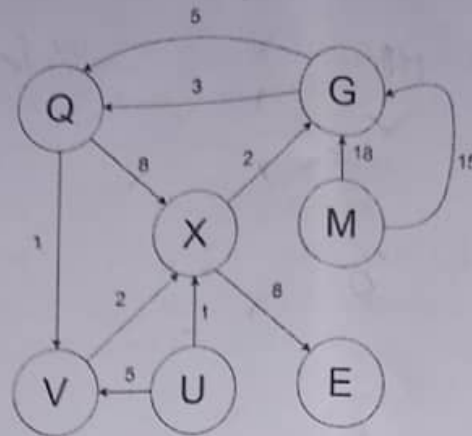
2. (a) Find two different (if exists) Minimum Spanning Trees from the following graph using Prim's and Kruskal's algorithm. (8)
- (b) Write two differences between the Prim's and Kruskal's algorithms. (2)



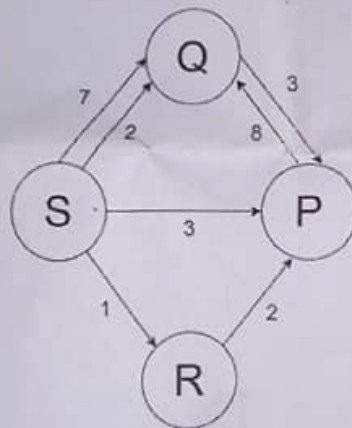
3. (a) Traverse the following graph in BFS and DFS order. Show the implementation process using stack or queue. Assume that U is the starting node. (8)
- (b) Suppose the same graph has both positive and negative weights. Which algorithm would you use to find the shortest path from the starting node U? Justify your answer. (2)



4. (a) Find the single-source shortest path distance from the following graph considering node M as the source. (7)
- (b) Which data structure does your selected algorithm use? Why is the data structure selection crucial to the algorithm's correctness? (3)



5. Find All Pair Shortest Path from the following graph. Show all steps. (10)



6. (a) Consider the following graph. Which algorithm would you use to find the shortest path from one node to all other nodes? Justify your answer. (2)
- (b) Answer the following questions: (6+2)
- Find the shortest distance from the source node A to all other nodes.
 - Using your own graph, prove that the Bellman-Ford algorithm doesn't work when a graph contains a negative weight cycle.

